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Portfolio Choice with Puts: Evidence from Variable Annuities

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This study investigated the asset allocation behavior of individuals who select an out-of-the-money long-dated longevity-put option on their investment funds. The asset allocations of these people within their variable annuity subaccounts are 5–30 percent more risky than the allocations of those who do not choose this protection. Investors who do not choose the longevity-put option follow the classic life-cycle, age-phased reduction in equity. A rudimentary model of utility-maximizing behavior is suggested that justifies the increased allocation to risk as long as the investor understands the payoff structure of the longevity put and is willing and able to exercise the annuity option if and when it matures in the money.

Motivated by the increasing need to examine and interest in examining consumer/investor behavior on the household level, we investigated the actual portfolio choices and asset allocation behavior of individuals who acquire portfolio insurance in the form of an out-of-the-money long-dated longevity-put option on their investment funds. We compared their sub-account allocations with those of investors who do not elect to pay for this type of protection. The evidence comes from variable annuity policies in a unique database of nearly 1 million policyholders contributed by seven different insurance companies.

We also offer a rudimentary model of utility-maximizing behavior in the presence of this longevity-put option that, indeed, justifies the increased allocation to risky assets/equity as long as the investor understands the payoff structure of the longevity put and is willing and able to exercise the annuity option if and when it matures in the money. That individuals have these characteristics is debatable, of course, in light of the long-standing body of evidence—first documented by Modigliani (1986)—that individuals dislike annuitization, despite its welfare-enhancing properties. We obviously cannot know the level of understanding or comprehension of individuals who acquire these rather complicated puts, but we will present some evidence that the distribution channel has an impact on the risk exposure of individual investors.

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We believe that this study is the first to examine actual asset allocations within variable annuity policies. In early 2008, variable annuities were a \$1.5 trillion market in the United States, according to LIMRA International, and the market is expected to grow as aging Baby Boomers take control of their retirement assets and generate their own pensions. Our results—which essentially document a type of “moral hazard” behavior—have both policy and risk management implications.¹

The Variable Annuity

At first glance, variable annuities (VAs) appear to be close cousins of mutual funds, which bundle individual securities—stocks or bonds—into diversified units or trusts. But VAs are formally classified as insurance policies in addition to being registered as securities. Traditionally, they have provided tax-sheltered growth, and they embed a variety of guarantees on the account value that are payable upon death of the contract owner or annuitant.

As with all insurance policies, and in contrast to standard exchange-traded put and call options, insurance companies charge for the downside protection of VAs by deducting an ongoing fraction of assets (rather than an up-front fee). The premiums for these policies are on the order of 50–150 bps of assets of the base of the guarantee (in addition to the basic management fees that one would pay on any mutual fund, managed money account, or exchange-traded fund). This unique pricing feature differentiates these derivative securities from the kinds of derivatives that can be analyzed in a standard Black–Scholes–Merton approach in which the

option premiums are paid up front and in advance. This aspect is important later in the discussion of optimal allocations and is worth noting from a behavioral point of view. Another characteristic of these guarantees that makes their pricing a challenging task is that they have long (up to several decades) maturity periods. As demonstrated by Bakshi, Cao, and Chen (2000), the pricing and valuation of long-dated options are much more complicated than the pricing and valuation of conventional short-term options. This article, however, is definitely not about pricing and valuing these options; rather, it is about the observed behavior of individuals in the presence—and absence—of these options.

The latest (i.e., after the year 2000) generation of VA contracts—and the specific focus of this paper—consists of contracts that have been financially engineered to provide an assortment of lifetime income guarantees that are meant to protect the policyholder against what the industry calls the “sequence-of-returns risk” and “longevity risk.” The first phrase refers to the chance that a retirement portfolio from which cash is being withdrawn will suffer early losses because of the path that investment returns take over time, and the second phrase loosely refers to the risk that retirees will live beyond their planning horizons and outlive their assets. Thus, the common denominator of the new generation of “guaranteed living income benefit” (GLIB) insurance riders is that they contain an implicit put option on financial markets plus some form of longevity insurance akin to a pure life annuity. Of course, in the concept of put–call parity, they can also be viewed as call options to annuitize at some variable strike price.

The (anecdotal) sales pitch for these products revolves around the idea that these guarantees allow investors to take on more financial risk than they would if they did not have these guarantees—which is precisely the issue at the core of this article. We used a unique database of policyholder behavior supplied by LIMRA to find out whether investors do, indeed, increase their risk exposure when selecting the longevity put. We also examine the theoretical merits of the advice to increase equity risk exposure; we do so by deriving the optimal asset allocation—under a stylized model of these products—in the presence of these optional riders.

Within the overall universe of GLIBs, we focus specifically on a type of rider that is called in the industry a “guaranteed minimum income benefit” (GMIB) and that we call “the longevity put.” The exact mechanics of this option are explained analytically in the section titled “Investing with a Longevity Put,” but here, we wish to say that at its core, a longevity put is a derivative

that allows the holder of the account to annuitize the funds at a guaranteed rate, which then provides a guaranteed level of lifetime income.²

The longevity put can be selected (or not) when the VA policy is initially purchased. This rider gives the holder the ability to annuitize some minimally guaranteed amount at some contractually guaranteed rate. Thus, for example, if a \$10,000 premium is placed into a VA, the insurance company might guarantee that at least \$15,000 can be annuitized in 10 years. The purchase price (or annuity factor) would be specified in the contract; for example, \$20 per \$1 of lifetime income. So, essentially, this example contract guarantees a minimal life annuity of at least $\$15,000 / \$20 = \$750$ per year, and if the market value of the (subaccounts within the) VA is more than the guaranteed \$15,000 over 10 years, the policyholder can annuitize the (greater) market value at market annuity rates.

We do not want the reader to get lost in the minutiae of these insurance policies because we believe that our research fits squarely within the portfolio choice literature. We investigated actual retail asset allocations of individual investors in the presence of performance and investment guarantees.

Relevant Literature

A number of recent papers have extended the set of decisions included in the portfolio choice problem to highlight the interaction of economic factors that the household, broadly defined, must consider and the risks that the household faces. This literature gained traction after the American Finance Association Presidential Address by John Campbell in January 2006 (printed in Campbell 2006). Goetzmann (1993), Yao and Zhang (2004), and Cocco (2005) focused on the role of the housing portfolio; Campbell and Cocco (2003) focused on optimal mortgage choices. Cairns, Blake, and Dowd (2006) examined portfolio choice in defined-contribution pension plans, Sundaresan and Zapatero (1997) examined the role of defined-benefit pensions, and Dybvig and Liu (2004) as well as Bodie, Detemple, Otruba, and Walter (2004) modeled the impact on asset allocation and portfolio choice of flexible retirement dates. Jagannathan and Kocherlakota (1996) and Viceira (2001) stressed the impact of aging. Faig and Shum (2002) studied the demand for illiquid assets; Koo (1998) and Davis and Willen (2000) modeled the role of labor income. Dammon, Spatt, and Zhang (2001) focused their attention on capital gains and income taxes; Heaton and Lucas (2000a) focused on the role of background risk. Chen, Ibbotson, Milevsky, and Zhu (2006) examined the role of insurance over the life cycle within a

portfolio choice context. Others have gone back to the basics and extended portfolio choice models to include sophisticated (and realistic) processes for investment returns (as did Chacko and Viceira 2005) or time-varying and mean-reverting risk premiums (as did Kim and Omberg 1996 and Detemple, Garcia, and Rindisbacher 2003).

The common denominator in these papers—and a frequently debated question—is how portfolio allocations to risky assets (stocks) should evolve with investor age. Finance practitioners consistently recommend lowering the stock content of one's portfolio with increasing age. This attitude is at the heart of many recent life-cycle or life-style funds promoted by fund companies. A popular rule of thumb is to have the share of equities be equal to 100 minus the investor's age.

Academic financial economists have turned to this issue mostly in the past decade. Several models of portfolio choice that include human capital, by and large, support the recommendation to decrease equity in the portfolio with increasing age, although the generated profiles are not necessarily monotonic.³ Recall the conclusion of Merton (1969, 1971) that age should not matter in the absence of human capital considerations.

A number of papers established some empirical facts. Several studies investigated the *determinants* of the stock share in investors' portfolios—in particular, the age effect. The fact relevant to this article is that most of these studies documented a profile in which the proportion of equity in the portfolio declined with increasing age.⁴ Other studies documented a hump-shaped pattern, with a declining part starting at the ages of 50–60.⁵

Keep in mind—especially in the context of the results we present later—that Ameriks and Zeldes (2004) showed that the age-to-equity profile is sensitive to model specifications. In fact, any regression explaining portfolio choice with age can include only two of the possible three variables: age, time, and cohort (age group). Although Ameriks and Zeldes found a hump-shaped relationship in a regression with age and time effects, they reported an increasing allocation to stocks/equity with age in a regression with age and cohort effects. The overwhelming majority of researchers, however, consider a cohort effect to be the least significant among the three and make the assumption that it is equal to zero. As a preview of some of our results, we also found a declining risk (equity, stock) allocation within the account as policyholders aged but we found that the age effect was substantially weakened when the insurance rider (i.e., the longevity put) was selected.

The common assumption of most of the aforementioned utility-based models is that investors act rationally. A growing body of literature documents, however, that investors do not always act rationally. Benartzi (2001) provided evidence that individuals put about one-third of their 401(k) pension accounts into their company's stock—a share that is far larger than would be justified by any asset allocation model. Benartzi and Thaler (2001) reported that many individual investors use a simple $1/n$ rule to allocate their assets among available choices regardless of the nature of those choices, which demonstrates that most individual investors do not understand such a basic investment principle as diversification. Moreover, Benartzi and Thaler (2002), based on their experimental results, argued that investors do not even have well-defined investment preferences. This evidence casts some doubt on whether annuitants really understand the benefits of the longevity put when they buy a VA contract.

VAs with longevity puts represent a type of portfolio insurance. Some studies on this subject (e.g., Brennan and Solanki 1981; Benninga and Blume 1985) have dismissed simple portfolio insurance as being suboptimal for a standard utility-maximizing investor. Other researchers, however, have sought to rationalize the widespread use of portfolio insurance in the financial industry. Leland (1980) argued that portfolio insurance can be beneficial for at least some individuals, and Brennan and Schwartz (1988) confirmed the optimality of time-invariant investment strategies.

The Data

The data that we used in our empirical analysis were provided by LIMRA International, which collected contract and product information from 10 member companies that sold VA policies in the January 2000–June 2004 period; the policies had to be in effect as of June 2004 and had to offer at least one type of longevity put at the time of purchase.

LIMRA's dataset has several advantages over most other sources of financial microlevel data. The two most popular data sources in household portfolio choice literature are the Survey of Consumer Finances (SCF) and the Panel Study of Income Dynamics (PSID). LIMRA's first advantage is the size of its dataset, which provides data on 812,367 individual VA contracts. In contrast, the PSID contains panel data on only about 8,000 families and the SCF has information on only 4,000+ households in each of its triennial surveys. Second, LIMRA's data come directly from insurance companies' original contracts and are thus

more accurate than self-reported information from the surveys. Third, LIMRA's data provide more detail with respect to asset allocation choices (e.g., amounts invested in small-, medium-, and large-capitalization equity styles, in international stock funds, and in investment-grade bonds versus high-yield bonds). Finally and most importantly, LIMRA's dataset is unique in terms of providing information on asset allocation within VAs with and without longevity puts. As Campbell (2006) rightfully argued, however, no perfect source of household financial data exists. Indeed, LIMRA's dataset lacks sufficient information about investors' personal characteristics and lacks data on their other investment accounts.

LIMRA's dataset contains 60 variables that provide information on investors and contract characteristics, such as investor age, gender, and state of residence; type of intermediary that sold the VA to the customer; the account value and its breakdown into amounts invested in several types of funds; and whether one of the three subcategories of longevity put was selected and the features of each type of guarantee.

We had to eliminate records with missing age information and records for which the values of the various subaccounts did not add up to the account value. The effect was to reduce our sample from 812,367 to 679,579 observations. We further limited our sample to those investors who (1) either selected no optional guarantee or (2) selected the GMIB (the longevity put) only. We did this for two reasons. First, compared with other types of optional guarantees, such as the guaranteed minimum withdrawal benefit or guaranteed minimum accumulation benefit, the longevity put (GMIB) is the closest equivalent to a life annuity with longevity insurance. Second, it is by far the most popular guarantee selected by individuals in our dataset. Of all the contracts with at least one guarantee selected, the contracts in which only the longevity put was selected represent 95 percent of the data.

As a result, we had a "clean" dataset of 660,336 observations with either a longevity put or no optional guarantees selected. These data served as the core of the subsequent analysis.

Analysis of the Empirical Results

We first consider the relationship between asset allocation and age for investors who selected the longevity put versus those who did not select any guarantees. Then, we discuss the results of our multivariate statistical analysis, which examined various factors that might explain the percentage of VA accounts invested in risky assets.

Asset Allocation, Age, and the Longevity

Put. We consider the percentage of VA accounts invested in high- and medium-risk (HMR) assets (that is, together, risky assets) as a function of investor age. We defined the following funds as HMR: large-cap, mid-cap, and small-cap stock funds; high-yield bond funds; balanced funds (combinations of equity and fixed income); specialty/sector funds; and international equity funds. We viewed investments in investment-grade bond funds, money market funds, and fixed accounts (accounts in which the insurance company guarantees an interest rate for a selected period of time) as low risk (LR).

Table 1 presents the percentage invested in HMR assets for 7 companies (out of the original 10) for 10 age groups in five-year buckets. (We excluded three companies with low numbers of policies that had a longevity put selected. The excluded companies represent 0.3 percent of all VAs with a longevity put.) We will focus our analysis on the two companies in our sample with the highest number of annuity policies, which we call Company 1 and Company 2. Combined, they represent 52 percent of VAs with a selected longevity put in our dataset. **Figure 1** presents the percentage allocation to risky assets for Company 1 (Panel A) and Company 2 (Panel B). In both panels, one age-versus-HMR line is for investors who selected a longevity put and the other is for those who did not select any additional guarantees.

As Panel A of Figure 1 shows, for Company 1, for investors who did not select any optional guarantee, the percentage invested in HMR assets declines with age, from 63 percent in the below-40 age group to 46 percent in the above-80 age group. This pattern is very different from the age-to-HMR profile of those who selected a longevity put. Among those investors, the percentage invested in HMR assets changed hardly at all with age; it stayed between 67 percent and 68 percent. Overall, the exposure of this group to HMR assets is significantly higher than that of the group who did not select any optional guarantee. We conclude that additional risk exposure in the group of longevity-put selectors is 5–18 percent of their VA account value.

The results for Company 2 demonstrate a similar trend. As Panel B of Figure 1 shows, the allocation to risky assets for the investors who selected a longevity put is substantially higher than it is for those who did not (10–31 percentage points higher) and declines with age much more slowly than for investors who did not select any optional guarantee. Interestingly, the share of assets invested in HMR assets is significantly lower in Company 1 than in Company 2 for any age group. Furthermore, in

Table 1. Age-Specific Asset Allocations across Seven Insurance Companies Offering VAs with a Longevity Put: Percentage in HMR Assets by Company

Age	C1		C2		C3		C7		C8		C9		C10	
	Without LP	With LP	Without LP	With LP	Without LP	With LP	Without LP	With LP	Without LP	With LP	Without LP	With LP	Without LP	With LP
< 40	63%	68%	76%	87%	88%	92%	77%	88%	47%	74%	85%	76%	86%	95%
41–45	62	68	73	85	87	92	75	87	46	75	83	75	83	95
46–50	62	68	71	85	88	92	73	85	41	73	84	78	81	94
51–55	60	68	67	84	88	92	71	85	38	71	82	76	75	93
56–60	55	68	62	82	88	91	68	82	33	70	79	74	70	92
61–65	51	67	59	82	86	90	65	81	29	66	78	73	66	92
66–70	50	67	58	82	86	89	64	80	27	64	76	73	64	91
71–75	49	67	59	82	83	88	66	78	26	70	75	68	63	90
76–80	51	68	62	80	83	85	64	78	27	na	73	na	60	88
> 80	na	na	47	78	80	87	na	na	na	na	na	na	na	na

na = not applicable.

Notes: Percentages invested in HMR assets for seven companies, C1–C3 and C7–C10, for 10 age groups in five-year buckets are presented. LP = longevity put. Companies 4, 5, and 6 were omitted.

Company 1, for investors who selected the longevity put, this share is very stable regardless of age. These differences could be attributable to different investment choices available in each company and to different asset allocation restrictions applied when a longevity-put rider is selected.

Figure 2 illustrates the relationship of the distribution channel to the selection of HMR assets versus LR assets. What we mean by “distribution channel” is the type of intermediary that sold the VA to the customer. Remember that most insurance companies that offer these products (especially the more complicated riders sold during the period of our study) do not sell directly to the end user, the individual investor. Thus, one requires an intermediary to purchase a VA. We found stark differences in the amounts allocated to HMR assets—both with and without the selection of the longevity put—depending on whether the VA was purchased through a bank, an independent financial adviser, a wirehouse broker, or some other channel. We do not mean to imply that any of the channels necessarily influenced or caused the allocation; the result could be a *cliente effect*. For example, people who are conservative, and, hence, likely to hold fewer HMR assets, may be more likely to purchase the VA through a bank and vice versa.

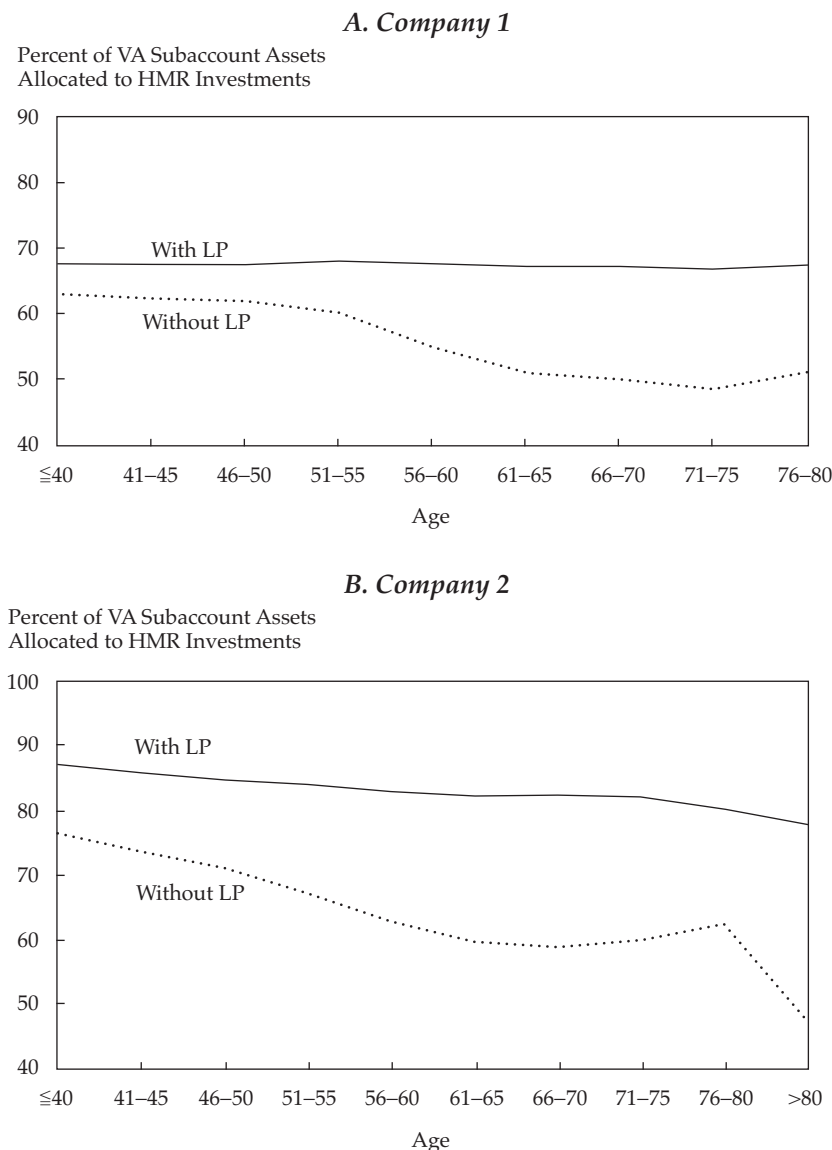
Multivariate Statistical Analysis. We examined whether the relationship between age and percentage invested in HMR assets observed in Table 1 holds in the presence of other variables that could explain HMR holdings. We used a set of linear models with the percentage invested in HMR

assets as the dependent variable, Y_i . We ran Model 1 for the whole sample, consisting of VA policies without any optional guarantees selected and those with a longevity put selected:

$$Y_i = \beta_0 + \beta_{LP} X_{LPi} + \beta_{AGE} X_{AGEi} + \beta_{AGE^2} X_{AGEi}^2 + \beta_M X_{Mi} + \beta_{ACC} X_{ACCi} + \beta_{IRA} X_{IRAi} + \sum_{j=1}^4 \beta_j^d X_{ji}^d + \sum_{k=1}^7 \beta_k^c X_{ki}^c + \varepsilon_i, \quad (1)$$

where i refers to the i th VA policy and the explanatory variables are as follows:

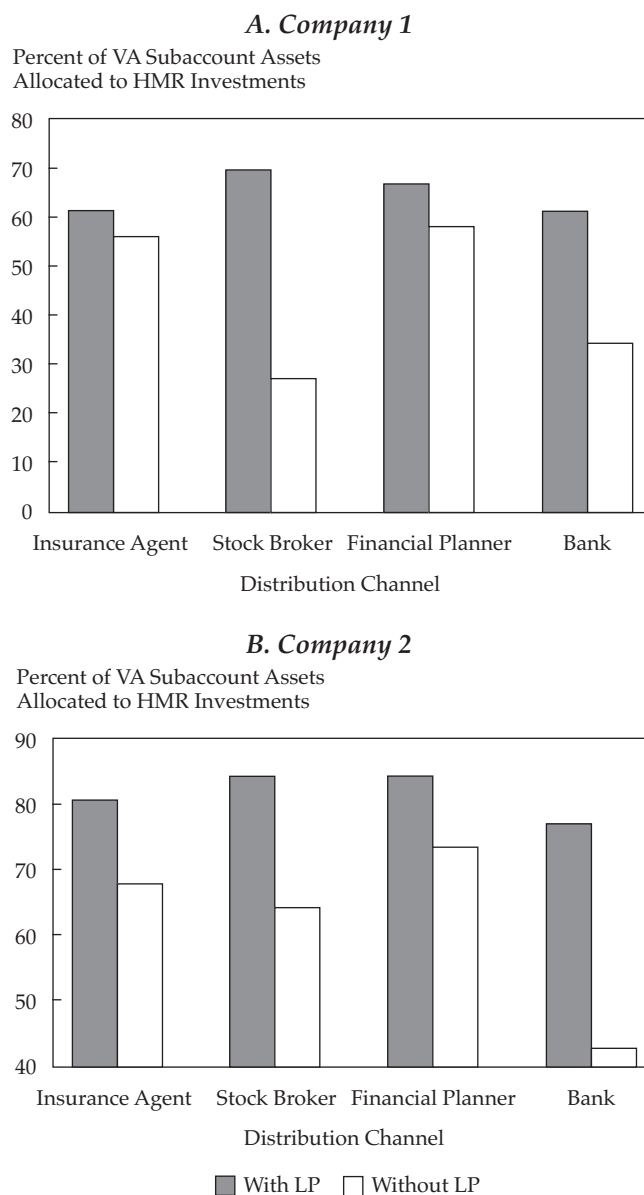
- X_{LP} = a dummy variable equal to 1 if the longevity put (LP) was selected and 0 otherwise.
- X_{AGE} = age.
- X_{AGE}^2 = age squared, to capture possible non-linearity in the age-to-HMR relationship.
- X_M = gender, a dummy variable equal to 1 when the annuity policyholder was a man and 0 when the policyholder was a woman to test the presence of a gender-based effect. A number of previous studies documented that men are generally considered to be more overconfident than women (e.g., Barber and Odean 2001) and may, therefore, invest proportionally more in risky assets than women do.
- X_{ACC} = log of account value. This variable can be viewed as a proxy for investors' wealth, although we are aware that we do not know the investors' other holdings, which could be substantial. Several studies have found that wealthier investors have more aggressive asset allocations (e.g., Heaton and Lucas 2000b; Shum and Faig 2006).

Figure 1. Allocation to Risky Assets as a Function of Age

Notes: The percentage of the variable annuity subaccount within the policy sold by each company that was allocated to HMR asset classes when the longevity put was selected and when no optional guarantee was selected is displayed. Funds that are regarded as HMR = large-cap, mid-cap, and small-cap equity funds; high-yield bond funds; balanced funds; specialty/sector funds; and international equity funds. The remainder of the available fund classes—investment-grade bond funds, money market funds, and fixed funds—are considered low-risk assets.

- X_{IRA} = a dummy variable to capture the tax status of the underlying account. The variable was equal to 1 if the VA was classified as an IRA (i.e., a personal pension plan) and 0 otherwise. Because investments inside IRAs are tax sheltered, we can expect to see a higher share of fixed-income assets in those accounts—the interest income of which is taxed at a higher rate than the rate on dividends or on capital gains on stocks. VAs are tax-sheltered investments already, so theoretically, where a VA is located should not make a difference in terms of the tax implications.
- X_{ji}^d = distribution channel, the intermediary through which the VA contract was purchased. We used four dummy variables, X_1^d through X_4^d , to represent the following distribution channels: insurance agent, stockbroker, financial planner, and bank employee. The omitted category is independent agent (because it was a

Figure 2. Allocation to Risky Assets as a Function of the Distribution Channel



Notes: The impact of distribution channel on allocation to HMR assets—with and without guarantees—is illustrated. The distribution channels for each company = career insurance agents, stock brokers (wirehouses), financial planners, and banks. The category of “independent agent” was omitted.

factor only in Company 7). This variable controls for the possible impact and influence of the intermediary on the percentage of risky assets selected. Some types of intermediaries may be better than others at explaining the value (or lack of value) of the longevity put, which would then influence asset allocation.

- X_{ki}^c = company. We controlled for the possibility that the share invested in risky assets was

company specific by using seven dummy variables, X_1^c through X_7^c , for the 7 companies (out of the original 10) that sold VA policies with LP riders. The omitted category consists of the three companies with the lowest number of VA contracts with a longevity put.

For Model 2, we used two subsamples. One subsample (A) is VA contracts for which no optional guarantee was selected; the second subsample (B) is

the contracts in which VA holders selected the longevity put. Model 2 differs from Model 1 by excluding the LP dummy variable:

$$Y_i = \beta_0 + \beta_{AGE} X_{AGEi} + \beta_{AGE^2} X_{AGEi}^2 + \beta_M X_{Mi} + \beta_{ACC} X_{ACCi} + \beta_{IRA} X_{IRAi} + \sum_{j=1}^4 \beta_j^d X_{ji}^d + \sum_{k=1}^7 \beta_k^c X_{ki}^c + \varepsilon_i. \quad (2)$$

The results of both models are presented in Table 2. The sole purpose of running Model 1 was to estimate the effect of the longevity put on the percentage invested in HMR assets. As can be judged by the positive and strongly significant coefficient on the LP variable, VA policyholders who selected the longevity put invested almost 14 percent more in HMR assets than those who did not select the longevity put. To assess whether regression coefficients overall are different for those who

did and those who did not select the longevity put, we performed the Chow test. The *F*-statistics from this test (not shown) prove that the differences are highly significant.

Next, we consider the results of Model 2. As judged by the negative coefficient on *AGE*, the share invested in risky assets declines with age in both subsamples. It declines much faster, however, for those investors who did not select any optional guarantee; the *AGE* coefficient in Model 2A is about three times higher in absolute value than it is in Model 2B, and based on the *t*-test, the difference is highly significant. Moreover, the *AGE*² coefficient is negative in Model 2A but positive in Model 2B. This result suggests that the share invested in risky assets is a concave function of age that declines faster for older ages for investors who did not select any optional guarantee. For those who selected the longevity put, the decline with age is much slower and a reversal even occurs for older ages.

Table 2. Linear Regression

Variable	Model 1:		Model 2			
	Whole Sample		A: Without LP		B: With LP	
	Estimate	<i>t</i> -Value	Estimate	<i>t</i> -Value	Estimate	<i>t</i> -Value
Intercept	0.8705*	188.97	0.9921*	131.92	0.9132*	59.22
Selected LP	0.1364*	159.76				
<i>AGE</i>	-0.0031*	-95.87	-0.0041*	-76.10	-0.0015*	-43.09
<i>AGE</i> ²	1.9E-07	0.72	-3.5E-06*	-8.02	1.8E-06*	5.57
Man	0.0185*	24.82	0.0256*	18.24	0.0123*	16.45
Log of account value	-0.0095*	-30.65	-0.0178*	-30.42	-0.0018*	-5.66
IRA/tax shelter	-0.0061*	-7.67	-0.0098*	-6.37	-0.0044*	-5.59
Distribution channel ^a						
Career agent	0.0280*	9.92	0.0757*	18.82	-0.0388*	-8.8
Stock broker	0.0362*	12.94	0.0296*	7.27	0.0112*	2.6
Financial planner	0.0598*	20.48	0.1474*	31.75	0.0025	0.57
Bank employee	-0.0970*	-33.65	-0.1433*	-34.42	-0.0576*	-13.02
Company ^b						
C1	-0.0963*	-33.86	-0.0928*	-23.29	-0.1331*	-9.26
C2	0.0671*	24.37	0.0808*	21.68	0.0383*	2.67
C3	0.1382*	45.94	0.2013*	38.11	0.0883*	6.14
C7	0.1009*	36.36	0.1654*	44.29	0.0285*	1.98
C8	-0.0988*	-29.89	-0.0450*	-9.91	-0.0489*	-3.33
C9	0.1061*	31.70	0.2315*	48.50	-0.0733*	-5.03
C10	0.1603*	54.02	0.2058*	47.94	0.1063*	7.38
Adjusted <i>R</i> ²	0.1980		0.1590		0.1540	
No. of obs.	640,570		272,564		368,005	

Note: Least-squares regressions with the percentage invested in HMR as the dependent variable are presented.

^aThe category of independent agent was omitted.

^bThe omitted category consists of Companies 4, 5, and 6.

*Significant at the 1 percent level.

In addition to age, several other variables are significant in explaining the percentage of an account invested in HMR assets in both models. As suspected, men invest more aggressively than women. A puzzling result is that VAs that are part of a tax-sheltered personal pension plan (IRA) have lower shares of risky assets and higher proportions of low-risk bonds. This result would be intuitively satisfying (i.e., bonds are more likely to be placed in tax shelters) were it not for the (perhaps not widely known) fact that VAs are already tax sheltered. Another puzzling result is that account value (wealth level) is negatively associated with an allocation to HMR assets.

The coefficients on the distribution channel dummy variables lead to the conclusion that for investors who did not select an optional guarantee, financial planners are associated with a higher allocation to risky assets than are other distribution channels. For those who selected the longevity put, however, relatively riskier asset allocation is associated with (and perhaps recommended by) stock brokers.

All of the company dummy variables are statistically significant. Because the companies are anonymous, we will not delve into the LP contract details (some contracts being more lucrative than others) that might drive this particular result. We did, however, repeat the regression analysis for all seven companies with contracts divided between those policyholders who chose the LP protection and those who did not. The results of these regressions are presented in **Table 3**.

As in the discussion of Figures 1 and 2, we focus this analysis on Company 1 and Company 2. In terms of the age-to-HMR relationship, these results are similar to those presented in Table 2. The share of risky assets declines with age much faster for investors who did not select a longevity put. Again, one can see some increase in HMR share in older ages for investors who did select it. Male investors generally allocate more to risky assets. In terms of account market value, however, the results are different depending on the company. In Company 1, the market value of the account is positively related to the share of risky assets, whereas in Company 2, the relationship is negative (similar to the result for the whole sample). Interestingly—and in contrast to the pooled results—in Company 1 and Company 2, the share invested in HMR assets is higher when the VA is in a tax-deferred account but only when the longevity put was *not* selected. When the longevity put was selected, the IRA status is associated with “more bonds” and fewer HMR assets.

Investing with a Longevity Put

We construct in this section a (simple) stylized model of portfolio choice in the presence of the LP (i.e., GMIB) option. Our objective is to derive a measure for the amount of extra risk an investor is willing to take when paying for a LP option that protects against downside risk.

We start by postulating a generic, unisex investor with initial wealth W_0 who is currently optimally allocated α^* percent to risky assets and $1 - \alpha^*$ percent to safe assets. These allocations take place within a VA but one without any riders. The risky and safe assets correspond to, respectively, the HMR and LR funds described earlier.

By selecting the (assumed optimal) allocation, the investor has revealed an explicit risk preference. And in the classic Merton (1969, 1971) model that assumes constant relative risk aversion (CRRA) and lognormal asset returns, the implied coefficient of relative risk aversion, γ , will be

$$\gamma = \frac{\mu - r}{\alpha^* \sigma^2}, \quad (3)$$

where

μ = expected return

σ = volatility

r = risk-free rate of return

For example, when the expected return from the HMR investment is $\mu = 0.09$, the volatility is $\sigma = 0.15$, and the risk-free rate is $r = 0.045$, an investor with a preexisting allocation of $\alpha^* = 0.50$ to risky (i.e., HMR) assets and $1 - \alpha^* = 0.50$ to safe (i.e., LR) assets reveals a risk aversion of exactly $\gamma = 4$ in a Mertonian Utopia.

Now, for this unisex (and rational) investor, we compute (solve for) how the investor might change the allocation to the HMR asset class if the investor were to be granted the longevity put. Specifically, let α^{**} denote the new (presumably optimal) allocation to the HMR asset class when the longevity put is wrapped around the investment account. The difference between α^{**} and α^* (the original optimal allocation in the absence of the put option) is the incremental allocation that is theoretically justifiable on the basis of the presence of the longevity put.

We will investigate the behavior of $\alpha^{**} - \alpha^*$ as a function of the various underlying contractual parameters, such as the strike price of the embedded LP option, the preexisting asset allocation (α^*), the investor's age, and the exogenous capital market parameter assumptions—namely, μ , σ , and r .

The LP guarantees the ability to convert or annuitize (in the worst-case scenario) the guaranteed amount, $W_0 e^{\eta T}$, at a prespecified rate, denoted g_x , in which η is a guaranteed investment return

Table 3. Company-Specific Regression Results

Company Variable	C1		C2		C3		C7		C8		C9		C10	
	Without LP	With LP	Without LP	With LP	Without LP	With LP	Without LP	With LP	Without LP	With LP	Without LP	With LP	Without LP	With LP
Intercept	0.5129*	0.6388*	0.9852*	0.9483*	0.8868*	0.9733*	1.0206*	0.9506*	0.9887*	0.8874*	1.0763*	0.8618*	1.2426*	0.9284*
AGE	-0.0026*	-0.0007*	-0.0045*	-0.0018*	-0.0014*	0.0014*	-0.0036*	-0.0024*	-0.0059*	-0.0034*	0.0029*	0.0016*	0.0066*	-0.0013*
AGE ²	-9.1E-06*	1.7E-06*	-9.3E-06*	3.8E-06*	4.2E-07	1.7E-07	5.2E-06*	5.0E-06*	-1.7E-06	1.4E-05*	3.1E-06*	3.2E-06	-1.5E-05*	-1.9E-06*
Man	0.0037	0.0036*	0.0364*	0.0203*	0.0227*	0.0075*	0.0251*	0.0187*	0.0262*	0.0297*	0.0336*	0.0222*	0.0283*	0.0067*
Log of account value	0.0052*	0.0030*	-0.0219*	-0.0067*	0.0023	0.0003	-0.0238*	-0.0060*	0.0244*	-0.0046	0.0115*	-0.0018	-0.0111*	0.0007
IRA/tax shelter	0.0079*	-0.0045*	0.0135*	-0.0051*	-0.0071	-0.0013	-0.0504*	-0.0035	0.0275*	0.0096	-0.0175*	-0.0241*	-0.0333*	-0.0025
Distribution channel														
Career agent	0.1780*	-0.0022	0.2169*	0.0268*			0.1674*	0.0519*						
Stockbroker	-0.0587*	0.0778*	0.2386*	0.0689*	0.0316*	0.0118*	0.1643*	0.0827*					0.1085*	0.0795*
Financial planner	0.2359*	0.0512*	0.3102*	0.0693*	0.0305*	0.0040								
Independent agent							0.2346*	0.0650*						
Adjusted R ²	0.1213	0.0174	0.1179	0.0184	0.0056	0.0051	0.0684	0.0264	0.0392	0.015	0.0291	0.0071	0.0678	0.0128
No. of obs.	36,165	89,947	65,855	103,348	10,915	57,627	75,490	41,730	25,112	6,219	16,234	9,534	27,152	59,346

Note: Least-squares regressions with percentage invested in the HMR as the dependent variable are presented.

*Significant at the 1 percent level.

and T is the contract horizon. Alternatively, of course, the investor can annuitize the account value $\tilde{W}_T$ at the (then) market rate, denoted by $\bar{a}_x$. The quantity $\tilde{W}_T$ is obviously unknown in advance and depends on both the selected allocations of the investor and the random performance of underlying market subaccounts. The subscript x on both annuity factors denotes the age at which the life annuity is priced or issued (e.g., age 70).⁶ Typical market values of $\bar{a}_x$ in the current interest rate environment might be \$10.20 at the age of $x = 70$, \$8.44 at the age of $x = 75$, and \$6.72 at the age of $x = 80$. This value is the cost of \$1 of annual lifetime income at these various purchase ages.

One can also think of g_x as the strike price of the LP option, although it is not a marketable put option that can be sold in the conventional sense but more of an exchange rate between a guaranteed amount and lifetime income. Either way, the LP option pays off, or promises, lifetime income, $\tilde{I}_x$, in the following amount:

$$\tilde{I}_x = \max \left(\frac{W_0 e^{\eta T}}{g_x}, \frac{\tilde{W}_T}{\bar{a}_x} \right). \quad (4)$$

The justification for Equation 4 is as follows: If the underlying market and subaccounts perform poorly (i.e., earn less than η per year during the T -year waiting period), the investor is guaranteed the ability to annuitize at the annuity factor rate of g_x . If the market value of the account at time T is greater than guaranteed amount $W_0 e^{\eta T}$, the investor can simply annuitize the (higher) account value at the (then) market annuity rate of $\bar{a}_x$. In fact, the investor does not even have to purchase this life annuity from the company that issued the longevity put. Investors can take their money anywhere and annuitize in the open market. Of course, all of this analysis assumes that the investor actually understands annuitization and wants to annuitize, which is a big and unexplored assumption here. (Note that $\tilde{W}_T$ is partially under the control of the investor and depends on the asset allocation vector, α , which is to be determined.)

We have now reached the issues of control and optimal portfolio choice. To convert the guaranteed lifetime income into a lump-sum value at horizon time T , we multiply guaranteed lifetime income $\tilde{I}_x$, as defined in Equation 4, by the (then) market annuity factor $\bar{a}_x$. The objective is to locate an asset allocation vector α^{**} that maximizes expected utility of wealth:

$$\alpha^{**} = \max_{\alpha} E \left[U \left(\tilde{I}_x \bar{a}_x \right) \right]. \quad (5)$$

The intuition is as follows: Imagine that there is a liquid secondary market for a lifetime income annuity. In theory, then, the annuitant could deannuitize income $\tilde{I}_x$ and obtain a lump sum in the amount of $\tilde{I}_x \bar{a}_x$. Thus, the true expected utility of wealth is as shown in Equation 5. Indeed, on the one hand, if the option expires out of the money and the market value of the lifetime income, $\tilde{W}_T$, is greater than the guaranteed amount of lifetime income, which is $\bar{a}_x (W_0 e^{\eta T} / g_x)$, the mark-to-market value is simply itself. On the other hand, if the option expires in the money, the guaranteed lifetime income will kick in. The objective, then, is to locate an asset allocation that maximizes the expected utility of this uniquely defined wealth. From the optimal timing of annuitization under this option, we can abstract the *utility of annuitization* per se.

Under the (new) optimal allocation, the expected return from the investment account will be

$$E(\text{Return}) = \alpha^{**} \mu + (1 - \alpha^{**}) r - f, \quad (6)$$

where

μ = expected return from the HMR funds

r = LR rate

f = an extra fee (i.e., insurance charge) for the longevity put

Remember that this put is paid for in periodic installments through deductions from the return in the account. In practice, the fee itself can be specified to be proportional to the value of assets in the account or can be proportional to the value of the guaranteed base. Either way, it reduces the growth rate of the investments.

We located the optimal allocation and the incremental justifiable risk allocations by simulation because an analytical approach is impossible and the numerical implementation is equally cumbersome. Specifically, we focused our computational attention on simulations for which the LP annuity factors g_x are in the vicinity of market annuity factors $\bar{a}_x$, which can be viewed as a fair LP case. We also generated a few scenarios in which the LP annuity factors were at a premium to (i.e., higher than) the market-based annuity rates, so $g_x > \bar{a}_x$. Also, for the majority of our simulations, we assumed that the guaranteed investment return, η , embedded in the longevity put was 6 percent per year.

We started with a 55-year-old investor who purchases a longevity put with a 15-year horizon. At the age of 70, the investor plans, with 100 percent certainty, to annuitize the account, either under the

terms of the guarantee (if the investment account earns less than 6 percent over the next 15 years) or at the market rates, generating lifetime income of $\tilde{I}_g$.

The results for the allocation that maximizes CRRA utility relative to various original allocations are in **Table 4**. We selected a CRRA utility function specification despite the fact that we are fully cognizant of the established fact that agents with CRRA preferences are not likely to purchase any form of portfolio insurance or investment protection. An almost classic result by now is that a CRRA utility maximizer (in continuous time) holds portfolios that are constantly rebalanced between the available

risky assets. We are not deriving a complete model of “who exactly decides to buy and pay for longevity puts.” Rather, we are supporting the (somewhat obvious) argument that “even” a CRRA investor will take on more risk when given the LP protection.

Several results from the extensive simulations are worth noting. The “positivity” of the $\alpha^{**} - \alpha^*$ variable—in other words, the justification for additional risk exposure—depends on the strike price of the option. Note, however, that the strike price in a longevity put is neither immediately obvious nor stated clearly within the contract or policy. Similarly, the guaranteed credited interest rate—which we denoted η —is in many cases quoted and used as

Table 4. Results of the Theoretical Allocation Model

Age at VA Purchase/Annuity Factor Rate, g_{age}	Original Percent Allocated to HMR Assets (α^*)	Implied γ	CE Wealth without LP	New Percent Allocated to HMR Assets (α^{**})	CE Wealth with LP
Age 55					
$g_{70} = 15$	30	6.67	\$1.42	100	\$2.28
	40	5	1.80	100	2.46
	50	4	2.14	100	2.65
$g_{70} = 20$	30	6.67	2.02	40–50	2.03
	40	5	1.88	90	2.19
	50	4	2.13	100	2.40
$g_{70} = 25$	30	6.67	2.02	30–40	2.02
	40	5	2.12	50–60	2.13
	50	4	2.18	80–90	2.29
Age 60					
$g_{70} = 15$	30	6.67	\$1.20	90–100	\$1.67
	40	5	1.47	100	1.78
	50	4	1.65	100	1.88
$g_{70} = 20$	30	6.67	1.60	30–40	1.60
	40	5	1.65	60	1.65
	50	4	1.65	80–100	1.75
$g_{70} = 25$	30	6.67	1.60	30–40	1.60
	40	5	1.66	50	1.66
	50	4	1.71	70	1.71
Age 65					
$g_{70} = 15$	30	6.67	\$1.26	30–40	\$1.26
	40	5	1.24	80–100	1.30
	50	4	1.28	100	1.34
$g_{70} = 20$	30	6.67	1.26	30	1.26
	40	5	1.29	50	1.29
	50	4	1.31	70	1.31
$g_{70} = 25$	30	6.67	1.26	30–40	1.26
	40	5	1.28	50	1.28
	50	4	1.31	60–70	1.31

Notes: Initial allocations to the risky assets assume that no optional guarantees are selected. CE (certainty-equivalent) wealth is based on \$1.00 of initial investment.

a simple interest rate. In other cases, the rate is compounded daily. Obviously, these dissimilarities make a difference to the optimal allocation as well as to the value of such a guarantee.

Our point here is more than simply that “the devil is in the details,” although that certainly is an important lesson. We wish to emphasize that there can be remarkable heterogeneity between various LP contract provisions even if they all seem to offer a 6 percent guaranteed return.

Moving on, we note that in the course of our extensive simulations to locate the optimal allocation, we observed a number of interesting results. First, in many cases, the optimal allocations were labeled “corner solutions.” Basically, if the LP annuity factor was favorable to the policyholder, in the solution, that policyholder took on as much risk as the contract allowed. In a sense, the policyholder purchased a put option and maximized the personal value of this option by taking on as much risk as possible. This allocation, however, assumed a 100 percent annuitization rate. If the contractual provisions were less favorable, which in our cases was denoted by a higher parameter value of g_v , the optimal allocations in the presence of the longevity put were no longer corner solutions. In some cases, the optimal share of risky assets when the investor selected the LP protection was only 5–10 percent higher than the one without the LP.

We caution readers that our simple model of optimal asset allocation in a longevity-put structure only scratches the surface of truly accurate representations of the dynamic control problem. Our objective in this section has not been to provide a complete theory of longevity-put option pricing and use. Further improvements would be to incorporate (1) the optimal timing of annuitization, (2) the stochasticity of interest rates, and (3) the credit risk of the insurer. We believe that these improvements would probably not change our main result: More risk is acceptable if the strike price of the embedded longevity put is sufficiently near the money—and if the policyholder understands what is involved in annuitization and wants to annuitize.

Conclusion

The topic of optimal asset allocation and portfolio choice over the human life cycle has been growing in scholarly and practical interest, especially since the passage of the Pension Protection Act of 2006 (PPA). Among other provisions, the PPA provides 401(k) plan sponsors with a “safe harbor” if they select an appropriate life-cycle fund as the default option for their employees. Hence, debate is ongoing about which life-cycle investment “prod-

ucts” are suitable and appropriate, including whether 401(k) plans should include annuities with lifetime income guarantees as default options. Indeed, an increasing number of normative and prescriptive economic models are being used to shed light on how individuals should invest life-cycle wealth as a function of their risk aversion, their age, and their other financial and human capital assets.⁷ To date, the empirical literature seems to provide conclusive evidence that young investors do hold more aggressive (riskier) equity allocations than older investors and that this proportion tends to decline over time as people approach retirement. Whether this behavior is optimal in the absence of human capital considerations continues to be the subject of debate. In this article, we started by confirming this stylized fact about asset allocation but looked at allocations from a unique perspective. We examined how actual allocations to risky assets change when individuals are able to select insurance/market protection in the form of longevity puts (i.e., options to annuitize). Specifically, we analyzed asset allocations inside VA policies—a type of tax-sheltered mutual fund in which optional insurance riders are available giving investors the right but not the obligation to annuitize at a contractual rate. We had data for close to 1 million policyholder accounts for policyholders spanning the ages 40 to 80. From these data, we could observe (1) their initial asset allocations and (2) whether they purchased and paid for a longevity-put option (at an additional fee of 50–80 bps of assets per year).

Our main finding is that individuals who acquire and pay for longevity-put options in their investment portfolios have 5–30 percent greater equity exposure than those who do not select the longevity put. Alas, to anyone trained in the valuation and pricing of American-style derivative securities, or even in the basic economics of moral hazard, this insight is virtually axiomatic. What is comforting is to see it confirmed on Main Street versus Wall Street. Just as importantly, from the insurance company’s risk management perspective, we provide evidence that one cannot make static assumptions about asset allocation and utilization. Moral hazard (or antiselection) is evident in all age groups, which is probably one reason that many companies have by now imposed asset allocation restrictions or do not allow more than 80–90 percent total equity exposure.

In addition, we illustrated that the observed numbers are consistent with the predictions of a simple model of portfolio choice in the presence of such guarantees. On a related note, anecdotal evidence suggests that compliance officers and

securities regulators have been expressing concern about inappropriate levels of investment risk taken by (elderly) investors. A common position taken by compliance departments at the broker/dealer wirehouses is that a so-called suitable asset allocation is intrinsic to the client and should not be “endogenized” by the selected riders. In contrast to this static view, our optimization model provides justification for the position that an insurance rider should affect investment allocations. In a sense, a given asset allocation cannot be declared suitable or unsuitable without knowing whether a longevity put was purchased.

As a corollary to our main results, we also confirmed that gender is a statistically significant determinant of asset allocation across the life cycle and that men tend to have greater equity exposure than women, which is consistent with a number of studies, including the widely cited work of Barber and Odean (2001). But then again, we found that in those accounts for which the longevity put was selected, women tended to behave more like men and older investors behaved as if they were younger.

We also documented a unique distribution channel effect—namely, that the intermediary selling the variable annuity policy is related to the amount of equity exposure within the policy as well as the likelihood of the longevity put being selected. This connection has not (yet) received much attention in the finance literature, but we hope our findings will spur further research into the behavioral impact of the sales intermediary on portfolio choice and asset allocation.

Finally, we documented that VA policies purchased within IRA tax shelters tend to have a greater allocation to bonds. This result is consistent with normative tax-based models, such as the work of Dammon, Spatt, and Zhang (2001), except that VAs are *already* tax sheltered. In a sense, therefore, our finding is somewhat puzzling. Why would a VA inside a tax shelter have more bonds than a VA outside a tax shelter? Perhaps, this practice is yet another behavioral puzzle.

We emphasize again that longevity-put options are not cash-back guarantees. Rather, they provide protection that is contingent on annuitization. In other words, the only way to exercise this

longevity put is to irreversibly annuitize the guaranteed amount, K , in exchange for lifetime income. In the language of option markets, if the underlying security ends up at a value of $S < K$, where the strike price is K , then the option’s payoff is $K - S$. However, this amount will be paid (by the insurance company) to the holder of the put only if the entire account value K is annuitized. It is not farfetched to assume that some investor-retirees might forgo the in-the-money value $K - S$ if their disutility from annuitization is large enough. And although Yaari (1965) himself would no doubt exercise this longevity-put option were it in the money by even a small amount, denoted by epsilon, whether the masses will do the same remains to be seen. This question is more than just an economic curiosity because it will affect pricing of the longevity put via this “lapsation factor.”

We conclude by noting that Modigliani (1986) dedicated a substantial part of his Nobel prize address to the so-called annuity puzzle, which is that most people shun voluntary annuitization despite its welfare-enhancing properties. Might this article be evidence that annuitization is palatable as a worst-case scenario? In other words, retirees may eschew immediate annuities but find comfort from acquiring (a much cheaper) immediate annuity that pays off only in a state of nature where their nest eggs have been decimated by poor market returns. Alternatively, this article might be classified as a contribution to the behavioral literature by documenting that individuals accept greater stock/equity market risk when they *believe* that they have downside protection.

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This article qualifies for 1 CE credit.

Notes

1. "Moral hazard" is the prospect that an individual insulated from risk may behave differently from the way he or she would behave if fully exposed to the risk.
2. The longevity put is closely related to a popular insurance rider called the "guaranteed minimum withdrawal benefit," another form of put option contained and selected within many VAs. See Milevsky (2006) and the actuarial references therein for an analysis of the guaranteed minimum withdrawal benefit and some of the risk management, pricing, and arbitrage issues related to it.
3. Examples of models of portfolio choice that include human capital are in Bodie, Merton, and Samuelson (1992), Jagannathan and Kocherlakota (1996), Cocco, Gomes, and Maenhout (2005), and Gomes and Michaelides (2005).
4. Studies that documented this age-to-equity profile include Bodie and Crane (1997), VanDerhei, Galer, Quick, and Rea (1999), Agnew, Balduzzi, and Sunden (2003), and Curcuru, Heaton, Lucas, and Moore (2006).
5. Examples of studies documenting a hump-shaped pattern include Yoo (1994), Bertaut and Starr-McCluer (2001), and Shum and Faig (2006).
6. We refer the interested reader to Milevsky (2006) for a detailed explanation of the actuarial pricing underlying the annuity.
7. See Ibbotson, Milevsky, Chen, and Zhu (2007) as an example of how portfolio choice interacts with insurance and annuity decisions.

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